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Using first principles calculations to identify new destabilized metal hydride reactions for reversible hydrogen storage†

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Received 8th December 2006, Accepted 6th February 2007
First published as an Advance Article on the web 26th February 2007
DOI: 10.1039/b617927d

Downloaded by Duke University on 25 July 2012
Published on 26 February 2007 on http://pubs.rsc.org | doi:10.1039/B617927D

Hydrides of period 2 and 3 elements are promising candidates for hydrogen storage, but typically have heats of reaction that are too high to be of use for fuel cell vehicles. Recent experimental work has focused on destabilizing metal hydrides through mixing metal hydrides with other compounds. A very large number of possible destabilized metal hydride reaction schemes exist, but the thermodynamic data required to assess the enthalpies of these reactions are not available in many cases. We have used density functional theory calculations to predict the reaction enthalpies for more than 300 destabilization reactions that have not previously been reported. The large majority of these reactions are predicted not to be useful for reversible hydrogen storage, having calculated reaction enthalpies that are either too high or too low, and hence these reactions need not be investigated experimentally. Our calculations also identify multiple promising reactions that have large enough hydrogen storage capacities to be useful in practical applications and have reaction thermodynamics that appear to be suitable for use in fuel cell vehicles and are therefore promising candidates for experimental work.

1. Introduction

Growing concern about the future availability and environmental acceptability of fossil energy has led to the search of alternative fuels. Hydrogen is one possible energy carrier that could one day replace direct combustion of fossil fuels. One of the most daunting challenges for widespread use of H_2 as a fuel is the absence of a commercially viable H_2 storage technology. A selection of relatively lightweight, low-cost, and high-capacity hydrogen storage devices must be available in a variety of sizes to meet different energy needs.[1,2] Current long-term US Department of Energy guidelines call for a system hydrogen storage density of 9 wt% and a volumetric density of 81 kg H_2 m^{-3} to enable fuel cell powered vehicles to replace petroleum fueled vehicles on a large scale.[3–5] Apart from these stringent density requirements, these systems should also preferably function at temperatures between ambient and 100 °C and at pressures below 100 bar.

It is unlikely these storage requirements can be met by any of the current hydrogen storage materials. High pressure tanks currently in use on prototype fuel cell vehicles, for example, do not meet the long-term volumetric and gravimetric density targets. New materials are therefore being investigated, including complex metal hydrides such as alanates, amides and borohydrides.[1,6] The discovery that Ti catalyst facilitates reversible storage of 5.6 wt% in $NaAlH_4$[7–9] was a significant advance in the area of complex hydrides. Many other complex hydrides have also been investigated, including lithium alanates,[10–12] magnesium alanates,[13] calcium alanates,[13,14] and potassium alanates.[15] Reactions involving amides and hydrides such as $LiNH_2$ and LiH have been shown to have a reversible storage capacity of 6.5 wt%.[16–20] Most hydrides of period 2 and 3 metals have relatively high hydrogen densities but are often thermodynamically very stable and do not typically release appreciable amounts of hydrogen until temperatures of 250 °C or higher are reached.[21,22] Such high temperatures are undesirable in a fuel cell vehicle because generating these temperatures incurs an unacceptable energy penalty. Conversely, some hydrides, such as AlH_3, have low thermodynamic stability and are therefore not directly reversible. To be a viable candidate for meeting the systems requirements outlined above, a hydride must have a reaction free energy that lies in a narrow range of values to allow reversible H_2 storage at practical temperatures and pressures. Therefore, a major challenge in developing new hydride-based storage materials is to identify materials with suitable reaction thermodynamics.

Practical hydrogen storage materials must not only exhibit favorable thermodynamic properties but also have sufficiently rapid kinetics of hydrogen charging and discharging. The work presented in this paper is limited to identifying materials for reversible H_2 storage with acceptable reaction thermodynamics. Once materials with acceptable storage capacity and thermodynamics have been found, other considerations such as reaction kinetics will become crucial in examining these materials for practical applications. Our focus on thermodynamics is motivated by the observation that the reaction

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kinetics of metal hydrides can, at least in principle, be significantly accelerated by using catalysts[7] or by controlling the particle size of reactants.[23,24] The best known example of acceleration of kinetics using catalysts is the demonstration by Bogdanovic and Schwickardi that small amounts of Ti greatly enhance the reaction kinetics associated with H_2 storage in $NaAlH_4$.[7] The mechanism of this catalysis remains a topic of intense debate.[7–9] Creating materials with smaller particle sizes appears not to have a significant effect on the overall system's thermodynamics except possibly in extreme situations such as in nanoclusters containing <40 metal atoms.[6]

Although the equilibrium between gaseous H_2 and a metal hydride is determined by both enthalpic and entropic contributions, enthalpy alone provides a useful means to screen candidate materials.[6] The temperature of dehydrogenation can be reliably estimated by $\Delta H = T\Delta S$, where ΔH and ΔS are the enthalpy and entropy change of the dehydrogenation reaction, respectively. Züttel *et al.*[25] have argued that the entropic contribution to metal hydride reactions is approximately 130 J K^{-1} mol^{-1} of H_2 for most simple metal-hydrogen systems. Lower values of ΔS can occur for complex metal hydrides. For example, for $LiBH_4 \rightarrow LiH + B + 3/2\ H_2$, ΔS is 97 J K^{-1} mol^{-1}.[26] If we estimate that $100 \leq \Delta S \leq 130$ J K^{-1} mol^{-1}, then for the dehydrogenation temperature to lie in the range 300–600 K requires that $30 \leq \Delta H \leq 78$ kJ mol^{-1} of H_2. Reactions with ΔH substantially higher than this range will not have appreciable hydrogen pressures until the temperature is unacceptably high. If ΔH is significantly less than 30 kJ mol^{-1} of H_2, the material will not be easily reversible. Based on this discussion, we pose the task of identifying metal hydrides of interest for reversible H_2 storage as finding systems for which $30 \leq \Delta H \leq 78$ kJ mol^{-1} H_2 and that yield significant gravimetric quantities of H_2.

The work of Reilly and Wiswall[27,28] showed that it is possible to modify the thermodynamics of hydrogenation/dehydrogenation reactions by using additives to form compounds or alloys in the dehydrogenated state that are energetically favorable with respect to the products of the reaction without additives. This concept is known as destabilization. The principle underlying this concept is that having a stabilized dehydrogenated state reduces the dehydrogenation enthalpy and therefore reduces the temperature required to generate a given pressure of hydrogen (or conversely increases the H_2 pressure at a given temperature). In recent years, several examples applying this concept to light metal hydrides have been studied. Vajo *et al.* introduced two schemes to destabilize MgH_2.[29,30] The destabilization of $LiNH_2$ using LiH was studied by Chen *et al.*[16,17] This material was shown to have a reversible capacity of 6.5 wt% H_2. Pinkerton *et al.*[31] and Aoki *et al.*[32] studied the destabilization of the $LiBH_4$ using $LiNH_2$, a combination of compounds that can release 11.9 wt% H_2 on completion. This reaction leads to the formation of an intermediate hydride recently identified as $Li_4BN_3H_{10}$,[33–39] dehydrogenation of which is exothermic at room temperature. Destabilization reactions of $Mg(NH_2)_2$ and LiH have also been studied.[40–44] Lu *et al.* have explored the destabilization reaction of $LiNH_2$ and Li_3AlH_6[45] and made an attempt to generalize the mechanism behind this process.[46]

The concept of destabilization relies on being able to identify compounds that can lead to formation of stabilized products, thus reducing the enthalpy of the overall reaction and increasing the partial pressure of the hydrides. This concept is very general and therefore a large number of potential destabilization reactions exist. The lack of thermodynamic data for many of the compounds that could potentially participate in destabilized metal hydride reaction schemes has limited the number of the reactions that have been considered. We have recently shown that plane wave density functional theory (DFT) can be used to estimate thermodynamic data for large numbers of possible compounds.[47–49] The aim of this article is to review how these calculations are performed and to describe their accuracy, then to describe how this approach has been used to identify promising destabilization schemes from a set of more than 300 potential reactions.

The theoretical underpinning of our results is described in section 2. The central goal in analyzing the thermodynamics of hydride decomposition reactions is to determine the reaction free energy at finite temperature. However, we argue that a computationally efficient strategy can be used that does not involve explicitly calculating this quantity for every conceivable reaction. Instead, calculations that only examine the enthalpy of reaction at 0 K and neglect zero point energies can be used to identify reactions that potentially have favorable thermodynamics. This approach eliminates many reactions from further consideration. For the remaining reactions, the finite temperature free energy of reaction is calculated by combining DFT total energy calculations with calculations for the phonon density of states for each solid material. The performance of a reaction of interest can then be characterized using either the finite temperature thermodynamic quantities or *via* a van't Hoff plot. The phonon density of state approach has also been used by researchers to analyze the structural stability of H-containing compounds and, more recently, to study the temperature dependent formation enthalpy of NaH.[50–52] The accuracy of this two-stage screening method is discussed in section 2.3 by comparing our calculated results with experimental data.

Our results for the thermodynamic properties of a large number of potential destabilization reactions are described in section 3. Our results for systems involving combinations of Al, B, Ca, N, H, Li, Mg, and Si, together with the theoretical methods that are necessary to describe the thermodynamics of destabilized metal hydrides, have been described earlier.[47,48] After reviewing these results, we present new results for destabilization schemes involving hydrides of Sc, Ti, and V and reactions that include C as a destabilizing agent. In all, we have examined more than 340 reactions. Although the great majority of these reactions will not be useful for reaching the H_2 storage goals defined above, our approach has identified multiple reactions with very promising thermodynamics.

We conclude in section 4 by discussing prospects for future work in this area, focusing especially on the limitations of our computational approach to this materials screening problem. The types of calculations we describe here should be viewed as a means of focusing and accelerating experimental studies, not as a means to replace experiments.

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2. Theoretical background and computational details

2.1 Theoretical approach

We are interested in characterizing the enthalpy of reaction and reaction equilibrium of metal hydride decomposition for reactions of the form

$$aA + bB + \ldots \leftrightarrow mM + nN + \ldots + H_2 \quad (1)$$

where all reactants and products except H_2 are solids. A complete way to approach this task is to compute the free energy of each component. The temperature-dependent free energy of each solid can be written as

$$G = U - TS + PV \cong U_0 + U_{vib}(T) - TS_{vib}(T) + PV \quad (2)$$

The first term on the right hand side is the electronic total energy of the material as computed using DFT. The vibrational contributions to the internal energy, U_{vib}, and to the entropy, S_{vib}, can be computed within the harmonic approximation once the phonon densities of states for the material are computed.[53] The free energy of gaseous H_2 was calculated using a combination of statistical mechanical and tabulated results. Treating gaseous H_2 as an ideal diatomic gas,

$$G_{H_2} = U_{0,H_2} + U_{trans+rot,H_2} + U_{vib,H_2} - TS_{H_2}(T) + (PV)_{H_2}. \quad (3)$$

The first term on the right hand side of eqn (3) is the DFT total energy of H_2. The translational, rotational and vibrational contributions to the internal energy are[54,55]

$$U_{trans+rot,H_2} = \frac{5}{2}RT \quad (4)$$

$$U_{vib,H_2} = \frac{N_A h\nu}{2} + \frac{N_A h\nu e^{-\beta h\nu}}{1 - e^{-\beta h\nu}} \quad (5)$$

where $\beta = 1/kT$, and k is the Boltzmann constant. The H_2 vibrational frequency, ν, was determined from a DFT calculation within the harmonic approximation. The zero-point energy for H_2 calculated in this way is 26.1 kJ mol^{-1}, in good agreement with the experimental value of 25.1 kJ mol^{-1}.[56] The entropy can be accurately evaluated using the Shomate equation with the coefficients specified in the NIST database at $P_0 = 1$ atm.[26] We treated H_2 as an ideal gas, so $PV = RT$.

Combining eqns (2) and (3), the free energy change associated with the forward reaction in eqn (1) is

$$\Delta G = \Delta G^* + G_{H_2} \quad (6)$$

where the star indicates that the free energy difference is calculated using only the solid phases. Because the volume change associated with generating gaseous H_2 is large relative to the volume changes in the solid components, PV contributions due to changes in the volumes of the solid phases are neglected. That is, we use $\Delta(PV)^* \cong 0$. Taking the activities of all solid phases to be unity, the equilibrium pressure of the overall reaction is then given by

$$\frac{P}{P_0} = \exp\left(\frac{-\Delta G(T)}{RT}\right) \quad (7)$$

Plotting the equilibrium pressures calculated from eqn (7) defines the van't Hoff plot for the decomposition reaction, eqn (1).[29,50,52,57]

Eqn (7) is well suited for performing detailed calculations for reactions of special interest. It is not ideally suited, however, for rapidly screening large numbers of potential reactions because the phonon density of states for each solid reactant or product can only be computed using computationally demanding methods. An approximate quantity that can be computed with much less effort is obtained by restricting attention to $T = 0$ K and neglecting zero point corrections for both the solid and gaseous species. This defines

$$\Delta U_0 = \sum_{\text{products}} U_0 - \sum_{\text{reactants}} U_0 \quad (8)$$

where U_0 is the total energy of one of the bulk structures of interest as calculated by DFT. Although the approximations made in constructing this quantity sound severe, this quantity in fact provides a useful estimate of the enthalpy change near room temperature. To see this, we first consider the free energy change for the overall reaction at $T = 0$ K,

$$\Delta G_0 = \Delta U_0 + \Delta U^*_{ZP} + U_{ZP,H_2}, \quad (9)$$

where the star indicates an energy difference determined using only the solids and the two final terms are zero point energies. The zero point energy of gaseous H_2, U_{ZP,H_2}, is 26.1 kJ mol^{-1} H_2. The change in zero point energy associated with the solid phases in all examples we have studied,[48,49] ΔU^*_{ZP}, is negative and has a larger magnitude than U_{ZP,H_2}. For example, ΔU^*_{ZP} is -35 kJ mol^{-1} H_2 for the direct decomposition of LiH and -43.7 kJ mol^{-1} H_2 for the reaction of $LiNH_2$ and MgH_2. We find that $-20 < \Delta U^*_{ZP} + U_{ZP,H_2} < -10$ kJ mol^{-1} H_2 for all the examples we have studied, with the direct decomposition reactions showing smaller differences than the reactions involving destabilized hydrides.[48]

The large energies associated with zero point effects are partially offset by the H_2 translational, rotational and PV contributions to the free energy change at finite temperatures. If we neglect PV contributions from the solid phases and finite temperature vibrational contributions to the enthalpy, we can approximate the finite temperature reaction enthalpy by

$$\Delta H(T) \approx \Delta U_0 + \Delta U^*_{ZP} + \Delta U_{ZP,H_2} + \frac{7}{2}RT. \quad (10)$$

The crucial point for our purposes is that the zero point energy contributions and the final term in this expression have opposite signs. Eqn (10) is not useful in itself because ΔU^*_{ZP} can only be established by computing the phonon density of states of the solid materials. Once this calculation is complete, $\Delta G(T)$ for the reaction of interest can be calculated directly from eqns (2) and (6). In terms of screening large number of reactions, however, eqn (10) suggests that we use

$$\Delta H(T) \approx \Delta U_0 + \delta E \quad (11)$$

where δE is empirically determined from the examples where both $\Delta H(T)$ and ΔU_0 are known. Examining data from a set of 6 reactions, we found that for $\Delta H(T)$ computed using DFT calculations at both $T = 300$ and 600 K, setting $-20 < \delta E <$

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0 kJ mol^{-1} H_2 makes eqn (11) accurate for the complete set of reactions, with the average value of δE being -10 kJ mol^{-1}.[48]

The approach described above has led us to a two stage screening strategy for considering large numbers of potential destabilized hydrides for H_2 storage. In the first stage, DFT total energy calculations are performed for all solid materials of interest, and finite temperature reaction enthalpies are estimated using eqn (11) setting $-20 < \delta E < 0$ kJ mol^{-1} H_2. If the range of reaction enthalpies predicted by this initial analysis does not overlap the range of enthalpies of practical interest at all, then the reaction is eliminated from further consideration. For the reactions that are not eliminated by this criterion, the phonon density of states is computed for each solid component in the reaction and the van't Hoff plot for the reaction is constructed as defined above.

2.2 Computational details

Plane wave DFT calculations were performed with the Vienna *ab-initio* simulation package (VASP).[58,59] The calculations examine a spatially infinite material using periodic boundary conditions in all the principal directions. We have used the projector augmented wave (PAW) method[60] for prediction of reaction enthalpies. Electron exchange and correlation effects were described using the generalized gradient approximation (GGA) with the Perdew-Wang 91 (PW91) functional.[61] Spin polarization was not used in any of our calculations. Most of the unit cells calculations were performed with a Monkhorst–Pack mesh of either $9 \times 9 \times 9$ or $8 \times 8 \times 8$ k-points. A smaller number of k-points were used for a few systems having the largest unit cells. Convergence of the energies with the number of k-points was checked for several test reactions (information in supplementary information of ref. 47). Geometry optimizations for bulk structures were performed by allowing all atomic positions and all cell parameters to vary. Geometric relaxations were carried out using a conjugate gradient algorithm until the forces on each of the unconstrained atoms were less than 0.03 eV Å^{-1}. For total energy calculations for reactions involving Li/B/N or their compounds, the energy cutoffs for all compounds were 260/325/425 eV. For all other reactions the energy cutoff was 260 eV. Accurate geometry optimizations within VASP require a reliable stress tensor, which was achieved by using an energy cutoff for geometry optimizations that was 30% larger than the cutoff for the final total energy calculations. Our calculations examined 152 compounds. For the majority of the compounds we examined, the initial structures for geometry relaxations were obtained from the tabulated experimental data available from Wyckoff,[62] Pearson's handbook[63] and the ICSD database.[64]

Calculations of the phonon density of states and the resulting vibrational contributions to the free energy were performed using the PHONON code developed by Parlinski.[65] These calculations used the same exchange–correlation functional as our total energy calculations and an energy cutoff of 425 eV. Force constants were computed using displacements of individual atoms of 0.03–0.07 Å. It is necessary to use a relatively large supercell to avoid interaction between images of the displaced atom when defining the phonon density of states.[53] In most cases we chose a supercell in such a way that it approximates a spherical shape. In these calculations, the number of k-points was chosen so that the density of k-points in k-space was approximately equal to the mesh used for our total energy calculations.

$LiBH_4$ undergoes a phase transition at 381 K from an orthorhombic phase to a hexagonal phase.[66] This phase transition temperature is at a value that is relevant for practical H_2 storage applications. The PHONON calculations for the high-T hexagonal phase indicate the presence of a large number of imaginary modes. This has also been seen in the calculations of Vegge *et al.*[52] Zarkevic *et al.*[67] have shown that the harmonic approximation in this case fails to account for the BH_4^- rotational (torsional) modes present in the high-T phase. We have taken into account the effect of this rotational mode explicitly in our calculations by equating the free energies of the two $LiBH_4$ phase at the experimentally observed phase transition temperature of 381 K. This is accomplished by expressing the Gibbs free energy of the orthorhombic and hexagonal phases as

$$G_{\mathrm{LiBH_4\text{-}ortho}} = G_{\mathrm{e}}^{\mathrm{ortho}} + G_{\mathrm{v}}^{\mathrm{ortho}}(T) \tag{12}$$

and

$$G_{\mathrm{LiBH_4\text{-}hex}} = G_{\mathrm{e}}^{\mathrm{hex}} + G_{\mathrm{v}}^{\mathrm{hex}}(T) + G_{\mathrm{tor}} \tag{13}$$

At 381 K the free energies of the two phases must be equal;

$$G_{\mathrm{e}}^{\mathrm{ortho}} + G_{\mathrm{v}}^{\mathrm{ortho}}(T) = G_{\mathrm{e}}^{\mathrm{hex}} + G_{\mathrm{v}}^{\mathrm{hex}}(T) + G_{\mathrm{tor}} \tag{14}$$

In the above expressions G_e is the electronic free energy of the system at 0 K and can be approximated by U_0, which is the total energy as calculated from VASP. $G_v(T)$ represents the vibrational free energy of the system as calculated from PHONON. The torsional free energy $G_{tor} = -RT(0.5 \ln T + C)$ has a form suggested by Zarkevic *et al.*[67] This gives $-RT(0.5 \ln T + C) = (U_0^{ortho} - U_0^{hex}) + [G_v^{ortho}(T) - G_v^{hex}(T)]$ at the phase transition temperature. Substituting $T = 381$ K and the other values, we have $C \approx 3.8$. We also note that $LiBH_4$ melts at 548 K.[68] Because all of our calculations are based on the solid state, the equilibrium H_2 pressures we predict for processes involving $LiBH_4$ above this temperature should be considered approximate.

2.3 Accuracy of DFT calculations

It is important to establish whether plane wave DFT calculations of the type described above can give quantitatively accurate information before they are used to screen large numbers of potential reactions. One view of this issue is provided by examination of Table 1, in which our DFT results are compared with experimental and tabulated thermodynamic data for six well known reactions. With the exception of the direct decomposition of MgH_2, the finite temperature reaction enthalpies in Table 1 derived from eqn (10) lie within ± 10 kJ mol^{-1} of H_2 of the experimental values; the deviation for the MgH_2 reaction is slightly larger than this. The results in Table 1 also provide a partial test of the validity of eqn (11) for estimating $\Delta H(T)$. A more detailed discussion of this point is presented in ref. 48.

Table 1 Reactions used to study the accuracy of DFT-based free energy calculations, with all energies listed in kJ mol^{-1} H_2. The amount of hydrogen generated when each reaction goes to completion is shown. Energy differences are listed for each reaction from experimental data[98,99] and thermodynamic tables[26] when available and from DFT total energy calculations. 'Tabulated' indicates that the reaction enthalpies have been calculated from the formation energies of the compounds available from the NIST database.[26] Finite temperature enthalpy predictions from current work are included for comparison

Reaction	wt% H_2	ΔH (Tabulated) (298 K)	ΔH (Expt.)	ΔH (DFT)	ΔU_0 (DFT)
$2\ LiH \rightarrow 2\ Li + H_2$	12.70	181.26	180.4 (298 K)	172.0 (298 K)	172.5
$MgH_2 \rightarrow Mg + H_2$	7.67	76.15	75.3 (298 K)	62.7 (298 K)	65.1
$3\ NaAlH_4 \rightarrow Na_3AlH_6 + 2\ Al + 3\ H_2$	3.74	—	38.4 (440 K)	27.8 (440 K)	31.9
$2\ Na_3AlH_6 \rightarrow 6\ NaH + 2\ Al + 3\ H_2$	2.75	—	47.4 (495 K)	45.1 (495 K)	50.9
$2\ MgH_2 + Si \rightarrow Mg_2Si + 2\ H_2$	5.00	37.24	—	35.5 (298 K)	37.9
$MgH_2 + 2\ LiBH_4 \rightarrow 2\ LiH + MgB_2 + 4\ H_2$	11.56	45.96	—	52.2 (298 K)	66.8

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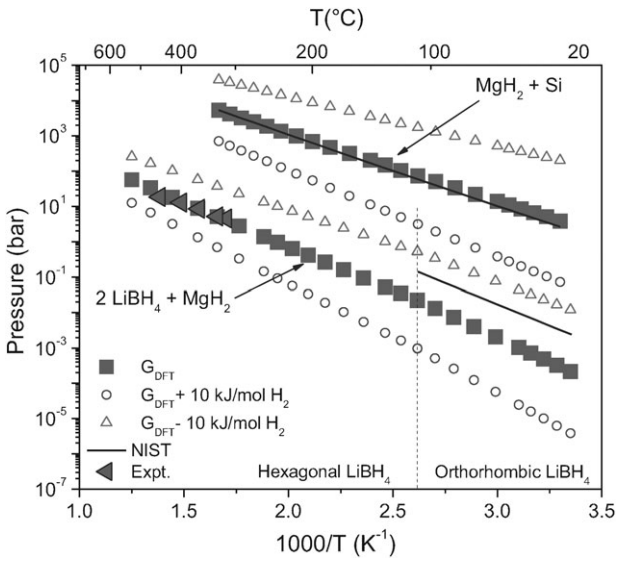

Fig. 1 Comparison of our calculations for $2\ LiBH_4 + MgH_2 \rightarrow 2\ LiH + MgB_2 + 4\ H_2$ (lower data set) and $MgH_2 + Si \rightarrow Mg_2Si + H_2$ (upper data set) with tabulated[26] and experimental data.[30]

A similar comparison can be made by examining the van't Hoff plots predicted by our calculations. One example is shown in Fig. 1. In this figure, the filled squares are the predictions of our DFT calculations, while the open symbols are the predictions made by simply adding ± 10 kJ mol^{-1} H_2 to our calculated $\Delta G(T)$. We have presented comparisons of this type for the other reactions listed in Table 1 previously.[48]

The comparisons above support the idea that our DFT calculations based on eqn (6) give an estimate of $\Delta G(T)$ that is accurate within about ± 10 kJ mol^{-1} of H_2. We readily admit that 10 kJ mol^{-1} can make a substantial difference in the practical performance of a real material, so precisely establishing the properties of the most promising materials experimentally is vital. For the purposes of screening large numbers of potential reactions to determine which of the many possible reactions are worth pursuing experimentally however, an uncertainty of ± 10 kJ mol^{-1} of H_2 is easily sufficient to select the small number of interesting reactions from the much larger collection of reactions with highly undesirable thermodynamic properties.

It is also worth noting within this discussion of accuracy that our DFT results are in good agreement with previous calculations of ΔU_0 by other groups.[32,69–73] A comparison with these previous DFT results is listed in Table 2. The results shown from previous calculations with $LiBH_4$ in Table 2 are all for the orthorhombic crystal structure.

3. Results

The central goal of our calculations was to examine a large number of possible destabilized reaction schemes for H_2 storage. To this end, we performed DFT calculations for 152 solid compounds with known crystal structures comprised from Al, B, C, Ca, N, H, Li, Mg, Sc, Si, Ti and V that are listed in Wyckoff,[62] Pearson's Handbook,[63] and the ICSD database.[64] The DFT-optimized structural parameters for these compounds are compared with experimental data in Table 3. The agreement between the DFT results and experiment is generally good, as expected. A few compounds that have significant van der Waals contribution need special mention. C (graphite) and $ScSi_2$ are examples of such layered compounds and it is not surprising that DFT does not make precise predictions for the structures of these materials. Similar results have been seen before for graphite and MoS_2.[74] The cohesive energies of these materials are dominated by the bonded interactions that are treated accurately by DFT, so

Table 2 Reaction enthalpies from our DFT calculations compared with values obtained using DFT by other researchers. ΔH and ΔU_0 values are listed in kJ mol^{-1} of H_2. The enthalpy for the reaction with the hexagonal and orthorhombic polymorphs of $LiBH_4$ is denoted by (h) and (o), respectively

Reaction	ΔH (other sources)	ΔU_0
$CaH_2 \rightarrow Ca + H_2$	181.5 [69]	175.2
$2\ AlH_3 \rightarrow 2\ Al + 3\ H_2$	10.04 [72]	11.1
$Ca(AlH_4)_2 \rightarrow 2\ Al + CaH_2 + 3\ H_2$	12.5 [71]	13.9
$LiBH_4 \rightarrow LiH + B + 1.5\ H_2$	71.3[70], 75 [73]	82.4 (o) 71.5 (h)
$LiNH_2 + 2\ LiH \rightarrow Li_3N + 2\ H_2$	99 [32]	109.4
$LiBH_4 + 2\ LiNH_2 \rightarrow Li_3BN_2 + 4\ H_2$	23 [32]	24.3 (o) 20.2 (h)

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Table 3 Comparison of the experimental and the DFT calculated structural parameters for the compounds in the reactions studied, with all distances in Å and angles in degrees. For $Ca(AlH_4)_2$, we compare our results with the previous DFT calculations of Løvvik.[71]

		Structural parameters/Å	
Compound	Space group	Experimental[62–64]	Calculated
Al	$Fm\bar{3}m$	a = 4.049	a = 4.05
$Al_{10}V$	$Fd\bar{3}m$	a = 14.492	a = 14.46
$Al_{12}Mg_{17}$	$I\bar{4}3m$	a = 10.524	a = 10.53
Al_2Ca	$Fd\bar{3}m$	a = 8.035	a = 8.035
$Al_2Ca_3Si_2$	$Immm$	a = 4.000, b = 18.240, c = 4.576	a = 4.09, b = 18.56, c = 4.55
Al_2CaSi_2	$P\bar{3}m1$	a = 4.130, c = 7.145	a = 4.16, c = 7.11
Al_2Li_3	$R\bar{3}m$	a = 4.508, c = 14.259	a = 4.45, c = 14.15
Al_2Sc	$Fd\bar{3}mS$	a = 7.58	a = 7.58
Al_2Ti	$Cmmm$	a = 12.0944, b = 3.3591, c = 4.0315	a = 12.14, b = 3.93, c = 4.00
Al_2Ti	$I4_1/amd$	a = 3.971, c = 24.313	a = 3.96, c = 24.33
Al_3Li	$Pm\bar{3}m$	a = 4.01	a = 4.03
Al_3Sc	$Pm\bar{3}m$	a = 4.105	a = 4.11
Al_3Ti	$I4/mmm$	a = 3.8537, c = 8.5839	a = 3.84, c = 8.61
Al_3V	$I4/mmm$	a = 3.780, c = 8.322	a = 3.76, c = 8.31
Al_4Li_9	$C2/m$	a = 19.1551, b = 4.4988, c = 5.4288, β = 107.361	a = 18.95, b = 4.47, c = 5.35, β = 107.43
AlB_2	$P6/mmm$	a = 3.009, c = 3.262	a = 3.01, c = 3.29
AlH_3	$R\bar{3}c$	a = 4.227, c = 11.244	a = 4.43, c = 11.85
AlLi	$Fd\bar{3}m$	a = 6.3757	a = 6.38
$AlLi_3N_2$	$Ia\bar{3}$	a = 9.461	a = 9.51
AlLiSi	$F\bar{4}3m$	a = 5.928	a = 5.95
$AlMg_4Si_6$	$C12/m1$	a = 16.53, b = 3.63, c = 6.72, β = 101	a = 16.37, b = 3.68, c = 6.73, β = 92.14
AlN	$P6_3mc$	a = 3.11, c = 4.98	a = 3.13, c = 5.01
AlSc	$Pm\bar{3}m$	a = 3.45	a = 3.37
$AlSc_2$	$P6_3/mmc$	a = 4.888, c = 6.166	a = 4.88, c = 6.14
$AlSc_2Si_2$	$P4/mbm$	a = 6.597, c = 3.3994	a = 6.60, c = 3.99
AlTi	$P4/mmm$	a = 2.832, c = 4.070	a = 2.80, c = 4.10
$AlTi_3$	$P6_3/mmc$	a = 5.764, c = 4.664	a = 5.72, c = 4.64
AlV_3	$Pm\bar{3}n$	a = 4.798	a = 4.78
B-α	$R\bar{3}mR$	a = 5.057, α = 58.06	a = 5.05, α = 58.08
B-β	$R\bar{3}mR$	a = 10.17, α = 65.12	a = 10.12, α = 65.27
C	$P6_3/mmc$	a = 2.464, c = 6.711	a = 2.47, c = 7.77
Ca	$Fm\bar{3}m$	a = 5.588	a = 5.57
$Ca(AlH_4)_2$	$Pbca$	a = 13.37, b = 9.28, c = 8.91 (ref. 71)	a = 13.14 b = 9.44, c = 8.86
Ca_2HN	$Fd\bar{3}m$	a = 10.13	a = 10.20
Ca_2LiSi_3	$Pnnm$	a = 11.24, b = 10.50, c = 4.39	a = 11.27, b = 10.54, c = 4.39
Ca_2N	$R\bar{3}m$	a = 3.638, c = 18.78	a = 3.64, c = 19.32
Ca_2N_3V	$C2/c$	a = 5.595, b = 10.410, c = 11.622, β = 92.342	a = 5.63, b = 10.48, c = 11.67, β = 92.04
Ca_3AlN_3	$P2_1/c$	a = 6.937, b = 6.149, c = 9.871, β = 94.01	a = 6.94, b = 6.16, c = 9.86, β = 94.01
Ca_3BN_3	$P4/mmm$	a = 3.5494, c = 8.2136	a = 3.57, c = 8.23
Ca_3N_2	$Ia\bar{3}$	a = 11.473	a = 11.54
Ca_3N_3V	$Cmcm$	a = 8.544, b = 10.38, c = 5.064	a = 8.58, b = 10.42, c = 5.08
Ca_5Si_3	$I4/mcm$	a = 7.64, c = 14.62	a = 7.67, c = 14.90
CaB_2C_2	$I4/mcm$	a = 5.3733, c = 7.4155	a = 5.36, c = 7.47
CaB_2C_2	$P\bar{4}2c$	a = 3.7921, c = 7.3966	a = 3.79, c = 7.62
CaB_6	$Pm\bar{3}m$	a = 4.145	a = 4.15
$CaCN_2$	$R\bar{3}mR$	a = 5.35, α = 40.47	a = 5.39, α = 40.57
CaH_2	$Pnma$	a = 5.925, b = 3.60, c = 6.838	a = 5.90, b = 3.57, c = 6.74
$CaLi_2$	$P6_3/mmc$	a = 6.268, c = 10.219	a = 6.20, c = 10.05
CaLiN	$Pnma$	a = 8.471, b = 3.676, c = 5.537	a = 8.55, b = 3.69, c = 5.58
$CaLiSi_2$	$Pnma$	a = 7.98, b = 3.80, c = 10.69	a = 7.96, b = 3.82, c = 10.66
CaMgSi	$Pnma$	a = 7.48, b = 4.40, c = 8.29	a = 7.49, b = 4.43, c = 8.33
$CaMg_2$	$P6_3/mmc$	a = 6.225, c = 10.18	a = 6.24, c = 10.08
$CaMg_2N_2$	$P\bar{3}m1$	a = 3.540, c = 6.0908	a = 3.55, c = 6.12
CaSi	$Cmcm$	A 4.559, b = 10.731, c = 3.890	a = 4.56, b = 10.79, c = 3.91
$CaSi_2$	$R\bar{3}mH$	a = 3.82, c = 15.98	a = 3.85, c = 15.89
Li	$Im\bar{3}m$	a = 3.510	a = 3.44
$Li_{12}Mg_3Si_4$	$I\bar{4}3d$	a = 10.688	a = 10.68
Li_2CN_2	$I4/mmm$	a = 3.687, c = 8.668	a = 3.71, c = 8.73
Li_2Si	$C2/m$	a = 7.7, b = 4.41, c = 6.56, β = 113.4	a = 7.54, b = 4.35, c = 6.64, β = 112.22
Li_3AlH_6	$R\bar{3}H$	a = 8.017, c = 9.455	a = 8.01, c = 9.46
Li_3BN_2	$P4_2/mnm$	a = 4.644, c = 5.259	a = 4.65, c = 5.26
Li_3N	$P6_3/mmc$	a = 3.552, c = 6.311	a = 3.56, c = 6.34
Li_3ScN_2	$Ia\bar{3}$	a = 10.045	a = 10.04
Li_4BNH_{10}	$I2_13$	a = 10.679	a = 10.65
Li_4NH	$I4_1/a$	a = 4.8765, c = 9.877	a = 4.87, c = 9.90
Li_5N_3Si (10 Si atoms)	$Ia\bar{3}$	a = 9.436	a = 9.435
Li_5N_3Si (11 Si atoms)	$Ia\bar{3}$	a = 9.436	a = 9.43, b = 9.42, c = 9.42
Li_7N_4V	$Pa\bar{3}$	a = 9.5859	a = 9.59

Table 3 (*continued*)

Compound	Space group	Structural parameters/Å	
		Experimental[62–64]	Calculated
Li_7N_4V	$P\bar{4}3a$	a = 9.6064	a = 9.61
Li_7N_4V	$P4_2/nmc$	a = 6.7575, c = 4.8818	a = 6.77, c = 4.89
Li_7Si_2	$Pbam$	a = 7.99, b = 15.21, c = 4.43	a = 8.05, b = 14.95, c = 4.48
$LiAlH_4$	$P2/c$	a = 4.845, b = 7.826, c = 7.917, β = 112.5	a = 4.83, b = 7.75, c = 7.87, β = 111.75
LiBC	$P6_3/mmc$	a = 2.7523, c = 7.058	a = 2.75, c = 7.056
$LiBH_4$	$Pnma$	a = 7.173, b = 4.434, c = 6.798	a = 7.13, b = 4.32, c = 6.57
$LiBH_4$	$P6_3mc$	a = 4.276, c = 6.948	a = 4.17, c = 7.26
LiCN	$Pnma$	a = 8.75, b = 3.74, c = 6.53	a = 8.80, b = 3.74, c = 6.57
LiH	$Fm\bar{3}m$	A 4.085	a = 4.026
LiMgN	$Pnma$	a = 7.1586, b = 3.5069, c = 5.0142	a = 7.19, b = 3.50, c = 5.02
LiN_3	$C2/m$	a = 5.627, b = 3.319, c = 4.979, β = 107.4	a = 5.64, b = 3.33, c = 4.99, β = 107.25
$LiNH_2$	$I\bar{4}$	a = 5.037, c = 10.278	a = 5.02, c = 10.26
LiSi	$I4_1/a$	a = 9.353, c = 5.743	a = 9.357, c = 5.75
Mg	$P6_3/mmc$	a = 3.209, c = 5.210	a = 3.23, c = 5.08
Mg(NCN)	$R\bar{3}mH$	a = 3.2734, c = 14.1282	a = 3.29, c = 14.17
Mg_2C_3	$Pnnm$	a = 6.4108, b = 5.2786, c = 3.7283	a = 6.43, b = 5.28, c = 3.74
Mg_2Si	$Fm\bar{3}m$	a = 6.390	a = 6.36
Mg_3BN_3	$P6_3/mmc$	a = 3.5445, c = 16.0353	a = 3.55, c = 16.12
Mg_3N_2	$Ia\bar{3}$	a = 9.972	a = 9.99
MgB_2	$P6/mmm$	a = 3.084, c = 3.522	a = 3.07, c = 3.52
MgB_2C_2	$Cmca$	a = 10.922, b = 9.461, c = 7.459	a = 10.91, b = 9.45, c = 7.45
MgB_4	$Pnma$	a = 5.464, b = 4.428, c = 7.472	a = 5.49, b = 4.39, c = 7.42
MgB_7 (Mg_8B_{56})	$Imma$	a = 5.970, b = 10.480, c = 8.125	a = 5.97, b = 10.47, c = 8.11
MgB_7 (Mg_7B_{56})	$Imma$	a = 5.970, b = 10.480, c = 8.125	a = 5.976, b = 10.49, c = 8.12
MgB_9N	$R\bar{3}mH$	a = 5.496, c = 20.0873	a = 5.49, c = 20.04
MgC_2	$P4_2/mnm$	a = 3.9342, c = 5.021	a = 3.93, c = 4.99
$MgCN_2$	$I\bar{4}2d$	a = 4.11, c = 7.562	a = 4.48, c = 6.19
MgH_2	$P4_2/mnm$	a = 4.517, c = 3.021	a = 4.47, c = 2.99
$MgSiN_2$	$Pna2_1$	a = 5.2725, b = 6.4733, c = 4.9862	a = 5.31, b = 6.49, c = 5.02
$N(SiH_3)_3$	$P\bar{1}$	a = 6.84, b = 8.17, c = 6.87, α = 95.5, β = 119.1, γ = 98.6	a = 7.14, b = 8.83, c = 7.18, α = 95.6, β = 119.0, γ = 99.2
N_4Si_3-α	$P31c$	a = 7.752, c = 5.619	a = 7.81, c = 5.66
N_4Si_3-β	$P6_3/m$	a = 7.602, c = 2.907	a = 7.66, c = 2.92
Na_3AlH_6	$P2_1/n$	a = 5.32, b = 5.56, c = 7.72, β = 89.72	a = 5.33, b = 5.53, c = 7.70, β = 89.63
$NaAlH_4$	$I4_1/a$	a = 4.99, c = 11.07	a = 4.99, c = 11.09
NaH	$Fm\bar{3}m$	a = 4.89	a = 4.81
$NaMgH_3$	$Pnma$	a = 5.4634, b = 7.703, c = 5.4108	a = 5.44, b = 7.63, c = 5.36
NaSi	$C2/c$	a = 12.19, b = 6.55, c = 11.18, β = 119	a = 12.15, b = 6.58, c = 11.05, β = 118.97
Sc	$P6_3/mmc$	a = 3.308, c = 5.2653	a = 3.29, c = 5.11
Sc_2BC_2	$I4/mmm$	a = 3.3259, c = 10.6741	a = 3.33, c = 10.66
Sc_2C	$P\bar{3}m1$	a = 3.387, c = 6.703	a = 3.31, c = 5.66
Sc_2C_3	$I\bar{4}3d$	a = 7.205	a = 7.67
$Sc_2V_3Si_4$	$P4_12_12$	a = 6.644, c = 12.27	a = 6.62, c = 12.25
Sc_3AlC	$Pm\bar{3}m$	a = 4.981	a = 4.50
Sc_3C_4	$P4/mnc$	a = 7.4873, c = 15.026	a = 7.49, c = 15.03
Sc_4C_3	$I\bar{4}3d$	a = 7.207	a = 7.19
Sc_5Si_3	$P6_3/mcm$	a = 7.861, c = 5.812	a = 7.87, c = 5.80
$ScAl_3C_3$	$P6_3mc$	a = 3.355, c = 16.776	a = 3.36, c = 16.77
ScB_{12}	$Fm\bar{3}m$	a = 7.402	a = 7.40
ScB_2	$P6/mmm$	a = 3.1478, c = 3.5169	a = 3.14, c = 3.52
ScB_2C	$P4_2/mbc$	a = 6.651, c = 6.763	a = 6.63, c = 6.76
ScB_2C_2	$Pbam$	a = 5.175, b = 10.075, c = 3.44	a = 5.28, b = 10.18, c = 3.43
ScC	$Fm\bar{3}m$	a = 4.51	a = 4.67
ScH_2	$Fm\bar{3}m$	a = 4.78	a = 4.75
ScN	$Fm\bar{3}m$	a = 4.44	a = 4.49
ScSi	$Cmcm$	a = 3.988, b = 9.882, c = 3.659	a = 3.98, b = 9.88, c = 3.67
$ScSi_2$	$P6/mmm$	a = 3.66, c = 3.87	a = 4.02, c = 3.62
Si	$Fd\bar{3}m$	a = 5.430	a = 5.46
SiB_6	$Pm3m$	a = 4.15	a = 4.13
Ti	$P6_3/mmc$	a = 2.994, c = 4.678	a = 2.92, c = 4.63
Ti_2AlN	$P6_3/mmc$	a = 2.994, c = 13.61	a = 3.00, c = 13.63
Ti_2C	$Fd\bar{3}m$	a = 8.6	a = 8.64
Ti_2N	$I4_1/amd$	a = 4.140, c = 8.805	a = 4.15, c = 8.89
Ti_2N	$P4_2/mnm$	a = 4.9452, c = 3.0342	a = 4.96, c = 3.04
Ti_3AlC	$Pm\bar{3}m$	a = 4.15	a = 4.18
Ti_3AlC_2	$P6_3/mmc$	a = 3.072, c = 18.73	a = 3.08, c = 18.63
Ti_3AlN	$Pm\bar{3}m$	a = 4.112	a = 4.13
Ti_3SiC_2	$P6_3/mmc$	a = 3.0654, c = 17.635	a = 3.08, c = 17.72
Ti_4AlN_3	$P6_3/mmc$	a = 2.991, c = 23.396	a = 2.99, c = 23.47
Ti_5Si_3	$P6_3/mcm$	a = 7.461, c = 5.1508	a = 7.47, c = 5.11

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Table 3 (*continued*)

Compound	Space group	Structural parameters/Å	
		Experimental[62–64]	Calculated
Ti_5Si_4	$P4_12_12$	a = 6.713, c = 12.171	a = 6.71, c = 12.18
Ti_8C_5	$R\bar{3}mH$	a = 6.115, c = 14.9	a = 6.14, c = 14.86
TiB	$F\bar{4}3m$	a = 4.202	a = 4.97
TiB	$Pnma$	a = 6.012, b = 3.06, c = 4.56	a = 6.11, b = 3.05, c = 4.57
TiB_2	$P6/mmm$	a = 3.031, c = 3.2377	a = 3.03, c = 3.23
TiC	$Fm\bar{3}m$	a = 4.328	a = 4.34
TiH	$P4_2/n$	a = 4.21, c = 4.60	a = 4.16, c = 59
TiH_2	$Fm\bar{3}m$	a = 4.5345	a = 4.42
TiN	$Fm\bar{3}m$	a = 4.2442	a = 4.25
TiSi	$Pnma$	a = 6.544, b = 3.638, c = 4.997	a = 6.54, b = 3.64, c = 5.01
$TiSi_2$	$Fddd$	a = 8.268, b = 4.8002, c = 8.5521	a = 8.27, b = 4.81, c = 8.56
V	$Im\bar{3}m$	a = 3.0258	a = 2.97
V_2B_3	$Cmcm$	a = 3.0599, b = 18.4290, c = 2.9839	a = 3.04, b = 18.41, c = 2.98
V_2C	$Pbcn$	a = 4.567, b = 5.744, c = 5.026	a = 4.54, b = 5.74, c = 5.03
V_2H	$I4_1/amd$	a = 6.0012, c = 6.6188	a = 5.86, c = 6.65
V_2H	Cm	a = 4.4566, b = 3.0022, c = 4.4760, β = 95.609	a = 4.43, b = 2.94, c = 4.43, β = 97.11
V_3B_2	$P4/mbm$	a = 5.755, c = 3.038	a = 5.71, c = 3.00
V_3Si	$Pm\bar{3}n$	a = 4.719	a = 4.68
V_6C_5	$P3_112$	a = 5.09, c = 14.4	a = 5.10, c = 14.36
V_6Si_5	$Ibam$	a = 15.966, b = 7.501, c = 4.858	a = 15.92, b = 7.48, c = 4.82
VB	$Cmcm$	a = 3.1, b = 8.17, c = 2.98	a = 3.05, b = 8.02, c = 2.96
VB_2	$P6/mmm$	a = 3.0, c = 3.06	a = 2.99, c = 3.04
VC	$Fm\bar{3}m$	a = 4.172	a = 4.16
VH_2	$Fm\bar{3}m$	a = 4.27	a = 4.19
VN	$Fm\bar{3}m$	a = 4.13711	a = 4.12
VSi_2	$P6_422$	a = 4.5726, c = 6.3744	a = 4.57, c = 6.37

these energies can be expected to be accurate even if the interlayer spacing is not.

The list of materials given in Table 3 enabled us to compute reaction enthalpies for 342 reactions for which these enthalpies were not previously known (data included in the ESI).† Fig. 2 shows the plot of the reaction enthalpies (−50 to 100 kJ mol^{-1} H_2) as a function of the wt% of H_2 that can be obtained from a particular reaction assuming the dehydrogenation reaction goes to completion. It is immediately clear that many of these reactions are uninteresting from the point of view of hydrogen storage for either thermodynamic or gravimetric reasons (or both).

We identify promising candidate reaction schemes by considering each of the reactions shown in Fig. 2 that can give an equilibrium pressure greater than 1 bar in the temperature range of 300–600 K based on the criterion defined in section 1 and at the same time yield >6.5 wt% H storage. Recognizing, as discussed above, that ΔU_0 provides only an approximation for ΔH, we adopted an inclusive strategy by including all reactions with $15 \leq \Delta U_0 \leq 75$ kJ mol^{-1} H_2 for further consideration. In Fig. 2a, reactions that can proceed in an alternative lower reaction enthalpy pathway have been excluded from our analysis as discussed below for reactions involving $Ca(AlH_4)_2$. The chosen gravimetric density cutoff of 6.5 wt% is slightly larger than the interim 2010 DOE targets.[3] We identify candidate reactions in an inclusive manner, consistent with our goal of motivating detailed experimental studies of potential candidates. While recognizing that this approach may yield some false positives, we feel it is more productive to list all the reactions identified in this way in detail rather than exclude examples that might be rejected by a more conservative approach. We reiterate that the reaction enthalpies of all the reactions we have considered are listed in the ESI,† so readers who prefer to apply a different screening criterion than the one we have suggested can do so.

3.1 Screening based on ΔU_0

Using the inclusive criterion for destabilized light metal hydrides we have previously identified seven promising reaction schemes.[47] The first one among these is the reaction already studied experimentally by Vajo *et al.*,[30]

$$2\,LiBH_4 + MgH_2 \rightarrow MgB_2 + 2\,LiH + 4H_2. \qquad (15)$$

This reaction can yield a maximum of 11.56 wt% H at completion, and has been demonstrated experimentally to yield >9 wt% H reversibly. Our 0 K DFT calculations indicate that this reaction has reaction enthalpy of 66.8 kJ mol^{-1} H_2 using the orthorhombic form of $LiBH_4$. A second reaction is the one suggested recently by Pinkerton *et al.*[31] and also by Aoki *et al.*:[32]

$$LiBH_4 + 2\,LiNH_2 \rightarrow Li_3BN_2 + 4\,H_2 \qquad (16)$$

This reaction yields 11.9 wt% H at completion, and is predicted by DFT to have ΔH = 24 kJ mol^{-1} H_2 using the PAW approach and the orthorhombic form of $LiBH_4$. Aoki *et al.* used USPP-GGA calculations to predict the enthalpy of dehydrogenation for the above reaction to be 23 kJ mol^{-1} H_2. The *p–c* isotherm measurements on the ball milled mixture of $LiBH_4$ and $LiNH_2$ at a temperature of 522 K indicated that 7.8 wt% H_2 was desorbed, corresponding to 66% of the total possible yield. Pinkerton and coworkers found that ball milling or heating the $LiBH_4$ + 2 $LiNH_2$ mixture causes it to react to form a new quaternary hydride phase with the

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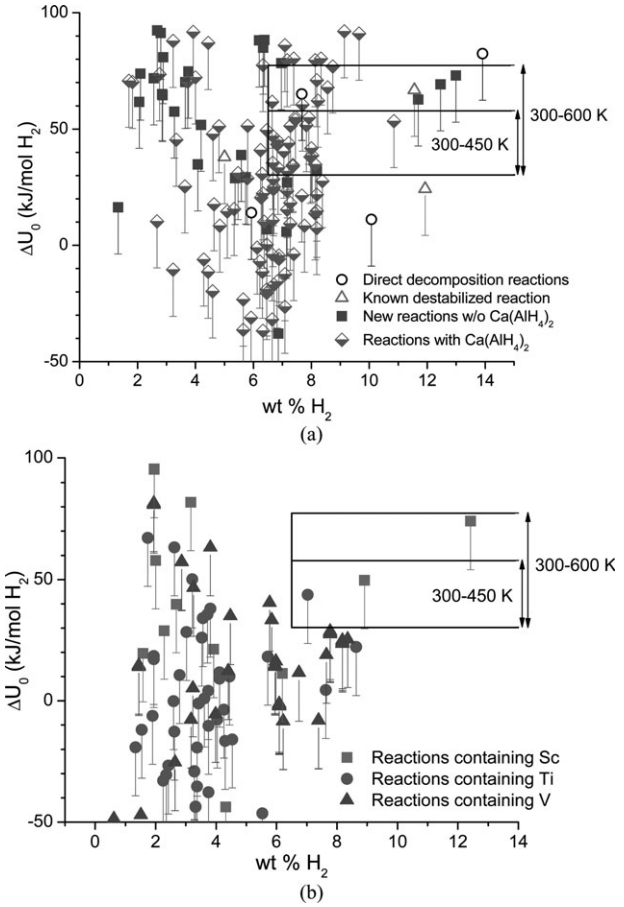

Fig. 2 Plots of ΔU_0 for reactions consisting of (a) B, Ca, H, Li, Mg, N and Si and (b) reactions including compounds of Sc, Ti and V. Both plots include approximate error bars indicating the difference between ΔU_0 and $\Delta H(T)$ for T in the range 300–600 K. The boxed areas indicate reactions which have >6.5 wt% H_2 and are likely to have $P_{eq} > 1$ bar in the indicated temperature ranges.

approximate composition $Li_3BN_2H_8$ (later corrected to $Li_4BN_3H_{10}$),[33] dehydrogenation of which was exothermic at room temperature. From these studies it is apparent that the reaction pathway identified in eqn (16) does not directly occur because of the formation of an intermediate quaternary hydride phase. This observation highlights an important limitation of our current computational approach, namely that we are limited to considering reactants and products with known crystal structures. Our approach cannot exclude the possibility that one or more of the new reactions we identify proceeds *via* a multi-step pathway involving reactants that are not included in the set of known structures we have considered. In such cases the thermodynamics of each decomposition step would be distinct from the overall reaction examined in our analysis.

Crucially, our approach also identifies multiple interesting reactions that had not previously been examined. All of these examples are summarized in Table 4. One example is

$$3\,LiNH_2 + 2\,LiH + Si \rightarrow Li_5N_3Si + 4\,H_2 \qquad (17)$$

This reaction yields 7.16 wt% H on completion. Our DFT calculations yield ΔU_0 = 23–34 kJ mol^{-1} H_2 using the PAW approach. The range of values for this reaction reflects the variants in the Li_5N_3Si crystal structure that were employed in our calculations.[47] A related reaction involves a mixture of $LiNH_2$ and MgH_2:

$$LiNH_2 + MgH_2 \rightarrow LiMgN + 2\,H_2 \qquad (18)$$

This reaction releases 8.19 wt% H on completion, with a DFT-predicted ΔU_0 of 31.9 kJ mol^{-1} H_2. The dehydrogenation properties of the mixture of $LiNH_2$ and MgH_2 in the molar ratio of 2 : 1 was studied by Luo.[75] The decomposition enthalpy for the 2 $LiNH_2$ + MgH_2 mixture was found to be 34 kJ mol^{-1} of H_2.

It has been observed from studies on the $Mg(NH_2)_2$ and LiH system that different products can be obtained depending upon the starting stoichiometry of the reactants, *e.g.*, Nakamori *et al.*[41] studied $Mg(NH_2)_2$ + 4 LiH → 1/3 Mg_3N_2 + 4/3 Li_3N + 4 H_2 and $Mg(NH_2)_2$ + 8/3 LiH → 1/3 Mg_3N_2 + 4/3 Li_2NH + 8/3 H_2 was studied by Leng *et al.*[42] and Ichikawa *et al.*[43] Nakamori *et al.*[41] have observed that increasing the LiH content in the mixture is an effective way to prevent ammonia release, however the trade off is that the amount of desorbed hydrogen is reduced. The observation that changing the molar ratios of the reactants gives rise to different products lends credence to our prediction that LiMgN may be obtained as a product of $LiNH_2$ and MgH_2, as given in (18) and also justifies the reactions given below in reactions (19) and (20).

Two other examples identified by our calculations are variants of (15):

$$4\,LiBH_4 + MgH_2 \rightarrow MgB_4 + 4\,LiH + 7\,H_2 \qquad (19)$$

$$7\,LiBH_4 + MgH_2 \rightarrow MgB_7 + 7\,LiH + 11.5\,H_2 \qquad (20)$$

These schemes differs from reaction (15) by including larger amounts of $LiBH_4$ as a reactant, forming MgB_4 or MgB_7 as a product rather than MgB_2. Reactions (19) and (20) release 12.5 and 13 wt% H on completion. Our DFT calculations predict that ΔU_0 is 69.2 (71.5–75.5) kJ mol^{-1} H_2 using the orthorhombic form of $LiBH_4$ for the reaction forming MgB_4 (MgB_7). The range of enthalpies for the latter reaction is due to the different structures of MgB_7 used to account for the vacancies in the crystal structure.

A fifth example involves CaH_2:

$$CaH_2 + 6\,LiBH_4 \rightarrow CaB_6 + 6\,LiH + 10\,H_2 \qquad (21)$$

Upon completion, this reaction releases 11.7 wt% H and the DFT-predicted ΔU_0 is 62.7 kJ mol^{-1} of H_2.

$Ca(AlH_4)_2$ has high enough gravimetric density to be used for hydrogen storage applications. Hence, many of the new reactions we considered in Fig. 2a involve $Ca(AlH_4)_2$. The predicted DFT reaction enthalpy for the decomposition of $Ca(AlH_4)_2$ *via*

$$Ca(AlH_4)_2 \rightarrow CaH_2 + 2\,Al + 3\,H_2 \qquad (22)$$

is only ΔU_0 = 14 kJ mol^{-1} H_2. Thus, even though many reactions with superficially attractive reaction enthalpies can be constructed using $Ca(AlH_4)_2$, they are all likely to be metastable pathways with respect to the direct decomposition reaction. Experimental studies for $Ca(AlH_4)_2$ dehydrogenation show that this compound does not decompose and release

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Table 4 Reactions with $15 \leq \Delta U_0 \leq 75$ kJ mol^{-1} H_2 and gravimetric densities >6.5 wt% (not involving $Ca(AlH_4)_2$)

Reaction	wt% H_2	ΔU_0
$3\ LiNH_2 + 2\ LiH + Si \rightarrow Li_5N_3Si + 4\ H_2$ (10/11 Si atoms)	7.16	34.2/23.3[a]
$LiBH_4 + 2\ LiNH_2 \rightarrow Li_3BN_2 + 4\ H_2$	11.9	24.3 (o) 20.2 (h)
$MgH_2 + 2\ LiBH_4 \rightarrow 2\ LiH + MgB_2 + 4\ H_2$	11.56	66.8 (o) 58.6 (h)
$LiNH_2 + MgH_2 \rightarrow LiMgN + 2\ H_2$	8.19	31.9
$4\ LiBH_4 + MgH_2 \rightarrow 4\ LiH + MgB_4 + 7\ H_2$	12.46	69.2 (o) 59.9 (h)
$7\ LiBH_4 + MgH_2 \rightarrow 7\ LiH + MgB_7 + 11.5\ H_2$	12.99	71.5–75.5 (o) 61.5–65.5 (h)
$CaH_2 + 6\ LiBH_4 \rightarrow CaB_6 + 6\ LiH + 10\ H_2$	11.69	62.7 (o) 52.9 (h)
$ScH_2 + 2\ LiBH_4 \rightarrow ScB_2 + 2\ LiH + 4\ H_2$	8.91	49.7 (o) 41.6 (h)
$TiH_2 + 2\ LiBH_4 \rightarrow 2\ LiH + TiB_2 + 4\ H_2$	8.63	22.2 (o) 14.1 (h)
$LiBH_4 + C \rightarrow LiBC + 2\ H_2$	11.95	45.1 (o) 37.0 (h)
$2\ LiNH_2 + C \rightarrow Li_2CN_2 + 2\ H_2$	6.97	31.4
$ScH_2 + 2\ LiBH_4 + C \rightarrow ScB_2C + 2\ LiH + 4\ H_2$	7.86	52.9 (o) 44.8 (h)
$ScH_2 + 2\ LiBH_4 + 2\ C \rightarrow ScB_2C_2 + 2\ LiH + 4\ H_2$	7.04	52.4 (o) 44.3 (h)
$MgH_2 + 2\ LiBH_4 + 2\ C \rightarrow MgB_2C_2 + 2\ LiH + 4\ H_2$	8.59	55.3 (o) 47.2 (h)
$CaH_2 + 2\ LiBH_4 + 2\ C \rightarrow CaB_2C_2 + 2\ LiH + 4\ H_2$	7.35	59.7 (o) 51.6 (h)

[a] Indicates results for final materials with varying Si loading. For reactions involving $LiBH_4$, the enthalpy for the reaction with the hexagonal and orthorhombic polymorphs of $LiBH_4$ is denoted by (h) and (o), respectively.

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H_2 until temperatures in excess of 200 °C are reached.[76] Even allowing for uncertainties in our DFT calculated reaction enthalpies the high decomposition temperature suggests that the decomposition of $Ca(AlH_4)_2$ is kinetically rather than thermodynamically limited. If reaction (22) is kinetically limited, it is at least conceivable that more facile reaction pathways with slightly higher reaction enthalpies may exist and be therefore experimentally observable. This rather speculative concept indicates that it is at least worthwhile identifying a number of these metastable reactions. Some of these $Ca(AlH_4)_2$ reactions that have reaction enthalpies in the range of 15–40 kJ mol^{-1} H_2 and release more than 6.5 wt% H_2 on completion have been listed in Table 5. It is important to reiterate that on purely thermodynamic grounds we do not expect these reactions to be observable. Nevertheless, experiments with $Ca(AlH_4)_2$ may be worthwhile as there is a small probability of identifying interesting metastable reactions.

3.1.1 Reactions including transition metal (Sc, Ti, V) hydrides. Hydrides of transition metals are usually not considered for hydrogen storage applications owing to their low gravimetric densities. We have applied our computational methods to reactions involving the hydrides of Sc, Ti, and V, and this process has yielded some interesting reactions. The hydrides of Sc are known to exist in the form of ScH_2 and ScH_3, but the structure of ScH_3 is not known exactly.[77] As a result, we have used the dihydride in our calculations. The hydrides of Ti exist as mono- and dihydrides.[78,79] For V, hydrides exist in the form of VH_2 and V_2H.[80–84] Experimentally, V_2H is known to exist in two different crystal structures.[82,83] We have considered both of these in our calculations.

The potential destabilization schemes we have considered involving hydrides of Sc, Ti and V using combinations of the materials listed in Table 3 are listed in the ESI.† It is no surprise that most of these reactions release relatively small amount of H_2 at completion, but there are several that involve more than 6.5 wt% H_2. Almost all the reactions involving ScH_2 we have considered hold little promise for high capacity reversible storage of H_2. The positive outcome of this screening exercise, however, is that two reactions are predicted to have favorable properties. The first of these is[49]

$$ScH_2 + 2\,LiBH_4 \rightarrow ScB_2 + 2\,LiH + 4\,H_2, \qquad (23)$$

a reaction that releases 8.91 wt% H_2 on completion. On a volumetric basis, this reaction stores 98.9 kg H_2 m^{-3}. This reaction is analogous to the destabilization of $LiBH_4$ with MgH_2 that has been studied experimentally by Vajo *et al.*[30] The calculated heat of reaction for eqn (23) is ΔU_0 = 49.7 kJ mol^{-1} H_2 based on *ortho*-$LiBH_4$.

The second promising reaction identified based on our initial calculations also involves ScH_2 and $LiBH_4$. In this case,

$$ScH_2 + 12\,LiBH_4 \rightarrow ScB_{12} + 12\,LiH + 19\,H_2, \qquad (24)$$

we calculated ΔU_0 = 74.1 kJ mol^{-1} H_2 using *ortho*-$LiBH_4$. This enthalpy is only marginally smaller than the DFT-computed enthalpy for direct decomposition of $LiBH_4$.[47,70,73] More importantly, this result suggests that these reaction products are more likely to initially react to form ScB_2 *via* eqn (23) until the ScH_2 is consumed. On this basis, we exclude the reaction forming ScB_{12} from further consideration.

Most reactions we have considered including hydrides of Ti also do not satisfy the gravimetric density requirements. Our DFT calculations indicate that direct decomposition of TiH_2 requires ΔU_0 = 142.7 kJ mol^{-1} H_2. Using TiH_2 to destabilize $LiBH_4$ can give rise to two types of reactions: the first one has TiB and LiH as products along with H_2 and the second one is again analogous to reaction studies by Vajo *et al.*[30] giving TiB_2 and LiH as products. For the reaction

$$TiH_2 + 2\,LiBH_4 \rightarrow 2\,LiH + TiB_2 + 4\,H_2 \qquad (25)$$

ΔU_0 using the orthorhombic form of LiBH4 is 22.2 kJ mol^{-1} H_2, while for the reaction

$$TiH_2 + LiBH_4 \rightarrow LiH + TiB + 2.5\,H_2 \qquad (26)$$

ΔU_0 is 43.7 and 162 kJ mol^{-1} H_2 for the orthorhombic and cubic forms of TiB, respectively. As reasoned above for the case of ScH_2 we suggest that the reaction of TiH_2 and $LiBH_4$

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would proceed *via* pathway of reaction (25). Using TiH as a reactant does not yield any promising reactions either.

Considering the reactions with hydrides of V with other elements and hydrides do not yield any interesting reactions. Although in this set of reactions there are a few which satisfy our screening gravimetric and thermodynamic requirements, we always also observed a lower energy pathway to form more stable alternative products. This is similar to the $Ca(AlH_4)_2$ systems discussed above. An example can be seen for the reaction

$$VH_2 + 4\,LiNH_2 + 3\,LiH \rightarrow Li_7N_4V + 6.5\,H_2. \quad (27)$$

A ΔU_0 value of $\sim$28 kJ mol^{-1} H_2 and gravimetric density of 7.78% makes it a possible candidate for further thermodynamic analysis. But our database of reactions suggests that VH_2 and $LiNH_2$ can also react *via* VH_2 + $LINH_2$ → VN + LiH + 1.5 H_2. This reaction has ΔU_0 of −5.4 kJ mol^{-1} H_2 which makes it thermodynamically more favorable than reaction (27). This indicates that all of VH_2 would initially react with $LiNH_2$ and we would be left with excess of $LiNH_2$ and LiH in the system if reaction (27) was attempted.

3.1.2 Reactions containing C as a destabilizing agent. Another possible destabilizing agent for metal hydrides is carbon. It can, of course, be argued that C can react with H_2 to form CH_4 or other hydrocarbons, but several studies have shown no formation of CH_4 in ball milled mixtures of C and metal hydrides.[24,85,86] Carbon in the form of graphite or alkali metal doped graphite have also been considered for H_2 storage.[87–89] Encouraged by this we have looked at reactions involving C for hydrogen storage applications that involve C in solid phase non-hydrocarbon species. Table 3 includes 31 C-containing solids of this kind.

Examining possible destabilization reactions that involve C-containing species has identified several interesting reactions. The first of these corresponds to the destabilization of $LiBH_4$:

$$LiBH_4 + C \rightarrow LiBC + 2\,H_2 \quad (28)$$

Our 0 K DFT calculations give ΔU_0 = 45.1 kJ mol^{-1} H_2 using the orthorhombic form of $LiBH_4$. This reaction yields 11.95 wt% H_2 at completion. A second reaction is the destabilization of $LiNH_2$:

$$2\,LiNH_2 + C \rightarrow Li_2CN_2 + 2\,H_2 \quad (29)$$

This reaction can provide >7 wt% hydrogen at completion. ΔU_0 from our calculations was 31.4 kJ mol^{-1} H_2 for this reaction.

The next two reactions correspond to the destabilization of $LiBH_4$ using both ScH_2 and C:

$$ScH_2 + 2\,LiBH_4 + C \rightarrow ScB_2C + 2\,LiH + 4\,H_2 \quad (30)$$

$$ScH_2 + 2\,LiBH_4 + 2\,C \rightarrow ScB_2C_2 + 2\,LiH + 4\,H_2 \quad (31)$$

These reactions have ΔU_0 of 52.9 and 52.4 kJ mol^{-1} H_2 and gravimetric densities of 7.9 and 7.0 wt% H, respectively. Comparing the above reactions with reaction (23) in which only ScH_2 and $LiBH_4$ react with a ΔU_0 of 49.7 kJ mol^{-1} H_2 we can see that the energies of all these reactions are very similar, indicating that these three reactions may compete when all three reactants are available. Adding C to the reaction does not change the 0 K reaction enthalpy, but may provide an alternative pathway for the reaction that may be kinetically favorable.

Similar to the reactions above, the next interesting reaction consists of adding C to the reaction (15) identified by Vajo *et al*. This reaction,

$$MgH_2 + 2\,LiBH_4 + 2\,C \rightarrow MgB_2C_2 + 2\,LiH + 4\,H_2, \quad (32)$$

has a gravimetric density of 8.6% and ΔU_0 of 55.3 kJ mol^{-1} H_2. This reaction enthalpy for (32) is lower than that for reaction (15) by 11.5 kJ mol^{-1} H_2 which indicates that, if feasible, the above reaction can provide a thermodynamically more favorable pathway.

The final interesting reaction we have identified using C as a destabilizing agent is similar to reaction (21) involving CaH_2 and $LiBH_4$,

$$CaH_2 + 2\,LiBH_4 + 2\,C \rightarrow CaB_2C_2 + 2\,LiH + 4\,H_2. \quad (33)$$

This reaction provide 7.35 wt% H_2 at completion. The 0 K reaction enthalpy using the orthorhombic form of $LiBH_4$ was 59.7 kJ mol^{-1} H_2 which is very similar to the 62.7 kJ mol^{-1} H_2 calculated for reaction (21).

As with all of our analysis, our predictions are purely thermodynamic, so they cannot give information about the kinetics of any reaction. Although reaction kinetics can potentially strongly limit any of the reactions we have discussed, it is plausible that these issues may be particularly vexing for reactions involving C because of the high barriers that exist to C–C bond breaking in graphite.

3.2 Free energy analysis of promising reactions

As described in section 2.1, our initial 0 K DFT calculations are a useful way to screen a large number of possible destabilization schemes relatively rapidly. Table 4 summarizes the most promising reactions identified with this approach from the >340 reactions we considered. For these promising reactions, it is useful to perform more complete calculations to describe the *T*-dependent free energy change associated with the reactions. We have performed these calculations for all of the reactions listed in Table 4 with two exceptions. The first exception is reaction 1 in Table 4. For this reaction one of the solid products has partial site occupancies by Li and Si. This situation would require a supercell with a very large number of atoms in order to examine reaction free energies. Unfortunately, calculations for such a large supercell are not currently feasible. The second exception is $LiBH_4$ + 2 $LiNH_2$ → Li_3BN_2 + 4 H_2. Experimental studies on this system have shown that this reaction proceeds in a two step manner, forming a quaternary hydride in the first step which in the second step undergoes exothermic decomposition to form the final products.[33–38] This example highlights an important limitation of our calculations, namely, that they are limited to materials for which the crystal structure is already known. We cannot exclude the possibility that some of the reactions we have analyzed may proceed in a multistep manner that involves compounds whose structures are not currently known. We return to this point in section 4.

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Table 5 Reactions involving $Ca(AlH_4)_2$ that have reaction enthalpies between 15–40 kJ mol^{-1} H_2 and can release more than 6.5 wt% H_2

Reaction	wt% H_2	ΔU_0
$3\ Ca(AlH_4)_2 + 2\ Si \rightarrow 2\ Al_2Ca + Al_2CaSi_2 + 12\ H_2$	6.69	24.1
$3\ Ca(AlH_4)_2 + 2\ Si \rightarrow 4\ Al + Al_2Ca_3Si_2 + 12\ H_2$	6.69	28.4
$6\ Ca(AlH_4)_2 + 17\ MgH_2 \rightarrow Al_{12}Mg_{17} + 6\ CaH_2 + 35\ H_2$	6.67	35.4
$Ca(AlH_4)_2 + 2\ LiH \rightarrow 2\ AlLi + CaH_2 + 4\ H_2$	6.85	33.3
$3\ Ca(AlH_4)_2 + 4\ LiH \rightarrow 4\ AlLi + Al_2Ca + 2\ CaH_2 + 12\ H_2$	7.17	31.8

Table 6 Enthalpic and entropic contributions to the reaction free energy of the destabilization reactions listed in Table 4. Energies are in kJ mol^{-1} H_2, entropies are in J K^{-1} mol^{-1} H_2. The equilibrium pressures are in bar. All results are from the DFT calculations defined in the text

Reaction	ΔU_0	ΔG_0	P_{eq} (300 K)	ΔH (300 K)	ΔS (300 K)	ΔH (600 K)	ΔS (600 K)
$MgH_2 + 2\ LiBH_4 \rightarrow 2\ LiH + MgB_2 + 4\ H_2$	66.8	47.3	2.7×10^{-4}	52.2	105.6	49.5	95.6
$LiNH_2 + MgH_2 \rightarrow LiMgN + 2\ H_2$	31.9	14.2	208.5	19.8	110.4	21.1	113.9
$4\ LiBH_4 + MgH_2 \rightarrow 4\ LiH + MgB_4 + 7\ H_2$	69.2	48.8	1.1×10^{-4}	53.5	102.5	50.0	88.7
$7\ LiBH_4 + MgH_2 \rightarrow 7\ LiH + MgB_7 + 11.5\ H_2$	71.5	50.7	5.2×10^{-5}	55.1	101.6	51.0	86.5
$6\ LiBH_4 + CaH_2 \rightarrow CaB_6 + 6\ LiH + 10\ H_2$	62.7	42.4	1.7×10^{-3}	47.0	103.7	43.05	90.0
$ScH_2 + 2\ LiBH_4 \rightarrow ScB_2 + 2\ LiH + 4\ H_2$	49.7	29.3	0.31	34.7	105.8	32.3	97.0
$TiH_2 + 2\ LiBH_4 \rightarrow 2\ LiH + TiB_2 + 4\ H_2$	22.2	1.3	2.1×10^{4}	6.5	104.4	4.2	95.7
$LiBH_4 + C \rightarrow LiBC + 2\ H_2$	43.6	25.2	0.90	31.8	105.1	28.9	95.0
$2\ LiNH_2 + C \rightarrow Li_2CN_2 + 2\ H_2$	31.4	6.1	5.2×10^{3}	11.2	108.4	12.4	111.3
$ScH_2 + 2\ LiBH_4 + C \rightarrow ScB_2C + 2\ LiH + 4\ H_2$	52.9	32.0	0.11	37.6	106.8	35.3	98.3
$ScH_2 + 2\ LiBH_4 + 2\ C \rightarrow ScB_2C_2 + 2\ LiH + 4\ H_2$	52.4	31.4	0.15	36.9	107.0	34.8	98.5
$MgH_2 + 2\ LiBH_4 + 2\ C \rightarrow MgB_2C_2 + 2\ LiH + 4\ H_2$	55.3	35.7	0.04	39.9	105.9	38.4	96
$CaH_2 + 2\ LiBH_4 + 2\ C \rightarrow CaB_2C_2 + 2\ LiH + 4\ H_2$	59.7	40.4	3.9×10^{-3}	45.9	106.9	43.3	97.5

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Table 6 lists the equilibrium pressures at 300 K for all the reactions for which we have performed the free energy calculations. The enthalpy and entropy changes for each of the reactions at 300 and 600 K are also listed in this table. One immediate observation from all these reactions is that the entropy change at 300 K lies in the narrow band of 100–110 J K^{-1} mol^{-1} H_2. These entropy changes lie within the bounds used to construct to regions with desirable material properties in Fig. 2.[48] If $\Delta S_{300\ K} = 105$ J K^{-1} mol^{-1} H_2, then to have $P_{eq} = 1$ bar at 300 K, $\Delta H_{300\ K}$ should be 31.5 kJ mol^{-1} H_2.

It is useful to split the reactions in Table 6 into three broad categories according to their predicted P_{eq} at 300 K. There are two reactions in the table that have high equilibrium pressures, with $P_{eq} > 1000$ bar at 300 K. These are the reactions for which $\Delta H_{300\ K} < 12$ kJ mol^{-1} H_2. Similarly, there are five reactions in the table with low predicted equilibrium pressures, specifically with $P_{eq} < 0.01$ bar at 300 K. For these reactions, $\Delta H_{300\ K} > 45$ kJ mol^{-1} H_2. Fortunately, there are six reactions in the table for which P_{eq} at 300 K is predicted to lie between 0.01 and 1000 bar. The van't Hoff plots for four of these reactions are shown in Fig. 3–6.

As we emphasized in section 2.3, it is appropriate to associate an empirical uncertainty estimate of ± 10 kJ mol^{-1} of H_2 with the free energies available from our DFT calculations. These uncertainties are shown in Fig. 3–6 using open symbols to bracket the DFT predicted equilibrium pressure shown as filled symbols. From eqn (7), this uncertainty corresponds to a range of pressures spanning ~ 3 orders of magnitude at room temperature. For considering practical applications, this is a large range of pressures. Unfortunately, there does not appear to be a straightforward way to make more precise predictions using plane wave DFT calculations with current functionals. The practical implication of this discussion is that while the six reactions in Table 6 that we identified above as having moderate equilibrium pressures appear the most promising of all the reactions we have studied, it is possible that some of the other reactions in Table 6 may also have useful equilibrium pressures. This issue can only be resolved *via* experimental studies. We hope that our calculations will motivate experiments of this type.

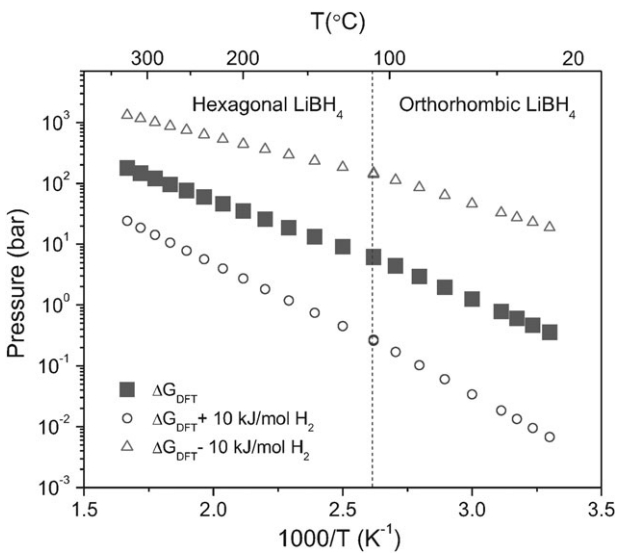

Fig. 3 van't Hoff plot from DFT calculations for $ScH_2 + 2\ LiBH_4 \rightarrow ScB_2 + 2\ LiH + 4\ H_2$. The open symbols represent estimates for the uncertainty in the calculations. For details, see the text.

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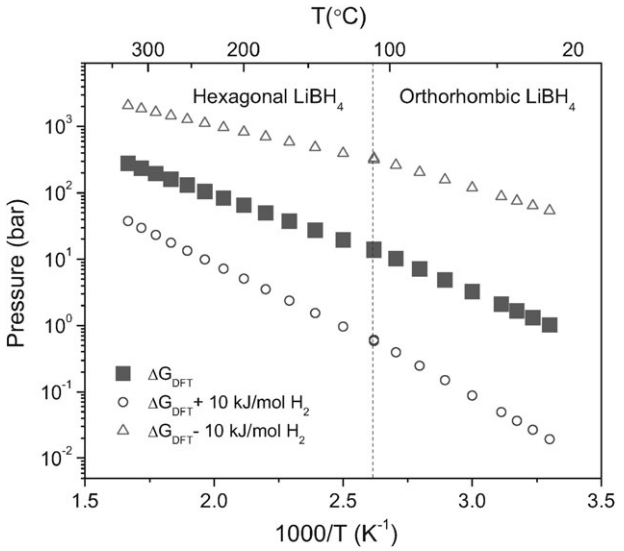

Fig. 4 van't Hoff plot for the destabilization reaction $LiBH_4 + C \rightarrow LiBC + 2\ H_2$ from DFT calculations.

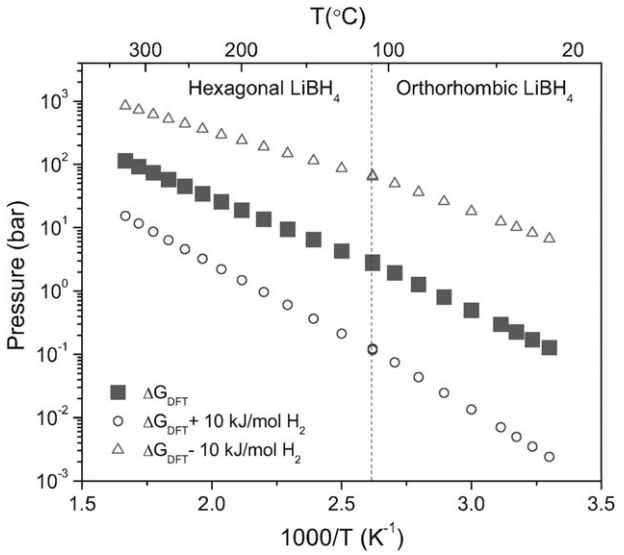

Fig. 5 van't Hoff plot from DFT calculations for $ScH_2 + 2\ LiBH_4 + C \rightarrow ScB_2C + 2\ LiH + 4\ H_2$.

4. Conclusions

We have used plane wave DFT calculations to examine the potential utility for hydrogen storage of a large set of destabilized metal hydride reactions for which thermodynamic data were previously unavailable. The intent of these calculations is to screen potential materials purely in terms of their reaction thermodynamics and storage capacity. These properties are useful for screening candidate materials because they are both well defined targets that can be considered as necessary (but not sufficient) conditions for a material to have practical use. One use of our calculations, therefore, is that we have identified a large number of candidate reactions that can be excluded from further consideration simply because they are far from reaching the applicable target for one or both of these

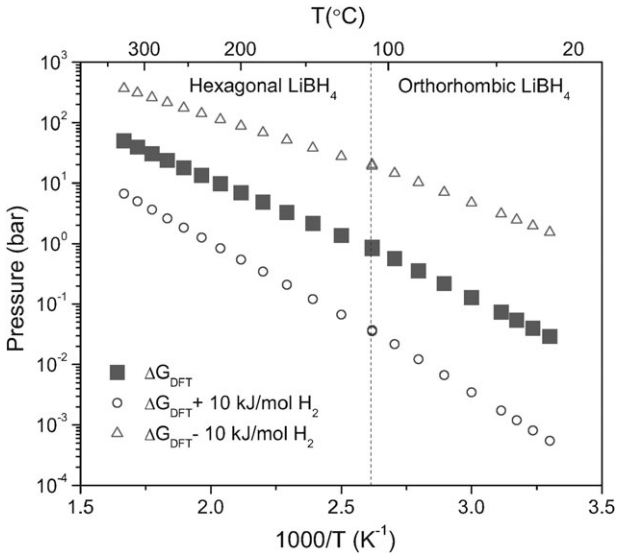

Fig. 6 van't Hoff plot from DFT calculations for $MgH_2 + 2\ LiBH_4 + 2\ C \rightarrow MgB_2C_2 + 2\ LiH + 4\ H_2$.

properties. More positively, our calculations have identified a number of reactions that have promising properties in terms of both their reaction enthalpies and their hydrogen storage capacity. These reactions are summarized in Tables 4 and 6.

To conclude, we discuss three important limitations of our calculations that need to be carefully considered in applications of our results or future similar calculations. The first and perhaps most obvious limitation is that DFT calculations are not exact representations of reality, so our predicted reaction free energies and enthalpies are not exact. Our calculations are performed in a self consistent way, using, for example, the same exchange–correlation functional for all calculations. But as with all DFT calculations, the detailed results from our calculations are dependent on the choice of functional. We feel that the most productive way to address this issue is to make direct comparisons between predicted and experimental results and to use these comparisons for establishing empirical uncertainties for our calculated quantities. Using this approach, we estimate that the uncertainty in our computed reaction free energies is ± 10 kJ mol^{-1} H_2. This means that at best we can expect to predict the equilibrium pressure for the reactions of interest only with a range spanning several orders of magnitude (see section 3.2). It would, of course, be helpful to be able to make more precise predictions than this, but this does not appear to be feasible with current quantum chemistry methods. Crucially, however, we argue that this level of accuracy is sufficient to judge between reactions that have potentially interesting properties and those that have little merit. As we have already stated, the purpose of our calculations is to motivate and focus experimental searches for destabilized hydrides. From this perspective, it is more useful to screen a large number of potential reactions with acceptable uncertainties than to perform highly accurate calculations that for reasons of computational expense must be limited to one or two reactions.

The second limitation of our approach is that we have only screened reactions that can be formed using solid compounds

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with known crystal structures. While this approach allows a large number of compounds to be examined (see Table 3), we can never exclude the possibility that a reaction will form a compound with a previously unknown crystal structure. We have already given one specific example of this situation. Our earlier calculations suggested that reaction of $LiNH_2$ and MgH_2 can produce LiMgN. This fact was indeed verified by Lu *et al.* who observed the formation of LiMgN between temperatures of 160 and 220 °C from their thermogravimetric analysis of $LiNH_2/MgH_2$ low energy milled mixture.[90] Hydrogenation of LiMgN did not, however, lead to the reverse of our proposed reaction. Instead, hydrogenation was observed to lead to formation of $Mg(NH_2)_2/MgH_2/LiH$. The intermediates in the reversible reaction that has been observed involving this mixture have been proposed to be $Li_2Mg(NH)_2$ or MgNH, the structures of which are unknown. While DFT calculations can play an important role in solving the structure of new compounds,[14,57,71,91–93] making predictions of solid state crystal structures in the absence of experimental information remains one of the outstanding open problems in computational materials science. Our calculations also did not consider the formation of gaseous species other than H_2. For some species of this type, notably NH_3, the presence of even trace levels of these gaseous products may be problematic in practical applications.

Our calculations also do not address hydrogen storage in the non-stoichiometric compounds such as metal alloys. This is clearly an interesting avenue to explore in order to consider the broadest possible suite of materials. Interesting results have been reported by Notten and co-workers on hydrogen storage in alloys of Mg and Sc.[94–96] In this case, the hydrides of MgSc alloys were found to have thermodynamic properties similar to MgH_2 with better charge transfer kinetics due to a Pd overcoat on the alloys. DFT calculations are well suited to describing the properties of H in ordered and disordered alloys,[97] so extensions of the kinds of calculations we have reported here should be useful for understanding possible applications of metal alloys in reversible hydrogen storage.

The two issues discussed above both relate to the ability of our calculations to correctly predict the reaction thermodynamics of a mixture of metal hydrides and other compounds. The final limitation of our approach is that the reaction thermodynamics is clearly not the only factor that will determine whether a material actually has practical value. Perhaps the most important physical issue associated with any reaction that has appropriate reaction thermodynamics and storage capacity is the reaction's kinetics. Our calculations provide no information about reaction kinetics, because they make no attempt to construct an explicit reaction path. Although the importance of reaction kinetics cannot be downplayed in the development of long term applications, it is appropriate to focus first on reaction thermodynamics because at least in principle it is possible to greatly alter reaction kinetics using catalysts, for example. It is clear from the large volume of work aimed at resolving the mechanism by which Ti catalyzes the reversibility of H_2 storage in sodium alanate[7–9] that understanding the details of how catalysts work for complex hydrides or destabilized hydrides will be a rich source of challenging problems for the foreseeable future. The task of understanding and ultimately controlling reaction kinetics in these systems will be crucial to efforts to move them towards large-scale practical applications.

Acknowledgements

This work was supported by the DOE grant number DE-FC36-05G015066 and performed in conjunction with the DOE Metal Hydride Center of Excellence. JKJ and DSS are NETL Faculty Fellows.

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